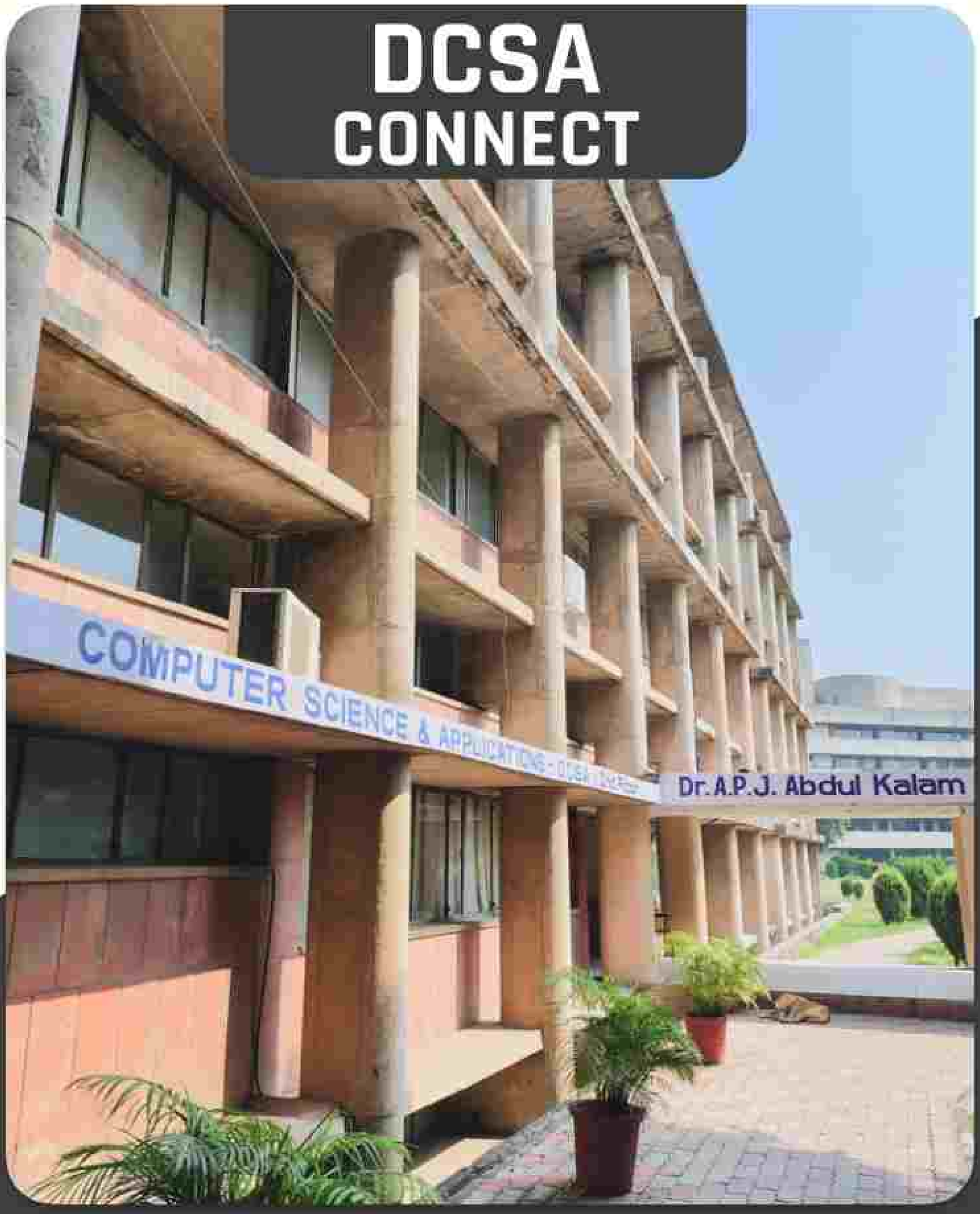
JAN-FEB 2025

EDITION

DCSA CONNECT



DEPARTMENT OF COMPUTER SCIENCE & APPLICATIONS,

PANJAB UNIVERSITY, CHANDIGARH

IN THIS EDITION

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TPC's Roadmap To Placement Success

On January 22, 2025, the Training and Placement Cell (TPC) held a meeting for all students to guide them through the placement process and its effective preparation. The session was led by Saakshat, the student coordinator of the TPC team. He explained the campus placement procedure and highlighted how the TPC supports students in securing job opportunities.

To further assist students in their placement preparation, he announced that specialized training sessions would be organized. These sessions will focus on clearing students' doubts and enhancing their skills in key areas essential for placements. Topics such as interview preparation, resume building, and communication skills will be covered, acquainting students with latest technology.

Additionally, the coordinator also highlighted that aptitude tests will be conducted for the students. These aptitude tests will assist the students during the recruitment process.

These tests will focus on logical reasoning, quantitative aptitude, and verbal ability, sharpening students' skills and improving their chances of success.



These tests will focus on logical reasoning, quantitative aptitude, and verbal ability, sharpening students' skills and improving their chances of success.

Saakshat emphasized the goal of achieving a high placement record for MCA and M.Sc. students at Panjab University's main campus and regional centers. He highlighted the commitment to fostering a supportive learning environment, enhancing students' academic and professional skills, and boosting their confidence for placement success.

He stressed a holistic approach, focusing on technical expertise, communication, problem-solving, and job readiness while encouraging participation in coding competitions.

Overall, the meeting was a significant step in preparing students for the upcoming placements, ensuring they have access to the necessary resources and guidance to thrive in their professional pursuits.

- Ishneet Kaur MCA-1(M)

Green Initiative: Environment Club Established



In alignment with the goals of the Swachh Bharat Abhiyan, an Environmental Club was officially formed in the department on January 22, 2025 with the purpose of promoting cleanliness and organizing environment related events within the department and the university.This initiative is a step towards fostering awareness about the importance of maintaining hygiene and environmental sustainability among students and staff. By addressing issues related to waste management, cleanliness, and greenery, the club aims to contribute meaningfully to the creation of a cleaner and healthier environment, both on campus and beyond.

The club is coordinated by Dr. Anuj Kumar and coordinated by Dr. Anuj Kumar and Dr. Kavita Taneja, whose leadership and guidance will help ensure the club achieves its objectives effectively. The club's core team comprises of a group of dynamic group of students who showed active participation. Sargundeep Singh (MCA- I (M)) and Amisha Sharma serve as Presidents, while Jatin Garg and Mehak Verma hold the positions of Vice-Presidents. Ishneet Kaur and Prabhat have been appointed as Secretaries, supported by Uday and Garima as Joint Secretaries. Managing the financial aspects are Parampreet Kaur and Mahak as Finance Secretaries. Ensuring the club's initiatives are well-publicized are the Media Coordinators, Jaskaran Singh and Nancy.

The Environmental Club has outlined several key activities to be implemented in the coming months. These include cleanliness drives in the department and campus, plantation campaigns to enhance greenery, and awareness programs to educate the university community about waste management and sustainable practices. With its well-structured leadership and enthusiastic members, the Environmental Club is poised to play a significant role in supporting the Swachh Bharat Abhiyan's mission and creating a lasting impact on the university's environment.

- Nancy MCA-1(E)

New GPU Workstation Installed in the Department

In Mid-January 2025, the department installed a high-performance GPU workstation to support computational tasks, simulations, and deep learning applications. The workstation, identified as HPZ4G5, is equipped with an NVIDIA 32GB graphics card, enhancing the department's research and computing capabilities.

This upgrade aims to improve efficiency in AI research, high-end rendering, and complex computations. It will be valuable resource for faculty, researchers, and students engaged in GPU-intensive projects, contributing to cutting-edge developments in various fields.

The addition of this workstation marks a significant step in strengthening the department's infrastructure for advanced computing and research activities.



- Nancy MCA-1(E)

Empowering Students: Personality Development & Soft Skills Workshop

The Personality Development and Soft Skills Workshop was a two-day event organized by the Central Placement in collaboration with the Training and Placement Cell, DCSA. It was held on February 3 and February 4, 2025, with the primary goal of helping students enhance their soft skills, an essential component of personal and professional growth. The workshop was attended by MCA and M.Sc. students of Chandigarh main campus and regional centres in online and offline mode to ensure broad participation.

The workshop was structured into four key sessions, each focusing on a vital aspect of skill development. The first session, 'Personality Development', was conducted by Dr. Manvinder Kaur, sharing strategies for improving self-awareness, confidence, and overall personal traits. Students were taught the importance of enhancing both their personality and interpersonal skills.

The second session, 'Communication Skills', was led by Dr. Amandeep Kahlon. This session emphasized on effective communication, covering both verbal and non-verbal techniques, active listening, and how to overcome common communication barriers.



The third session, 'Resume Writing', was conducted by Ms. Navneet Kalsi, providing essential insights on crafting a professional resume. Students learned how to highlight their skills, qualifications, and experience effectively to make a strong impression on potential employers.

The final session, 'Interview Skills', was led by Dr. Sanjiiv Gupta, sharing techniques to excel in job interviews. The session covered interview preparation, handling questions confidently, body language, and strategies for dealing with challenging interview scenarios.

After each session, students were encouraged to interact with the facilitators by asking questions and seeking clarification.This interactive approach allowed students to gain deeper insights and practical advice on applying the skills in real-life situations.

The Chief Patron of the event was Prof. Renu Vig, Vice Chancellor of Panjab University, with Prof. Meenu Sharma serving as the Honorary Director. The entire event was successfully concluded under the guidance of Dr. Anuj Sharma (Chairperson, DCSA), Dr. Anuj Kumar and Dr. Kavita Taneja (Placement Officers, DCSA).

In conclusion, the workshop was a significant success, providing students with valuable skills in personality development, communication, resume writing, and interview techniques. The students actively participated, gained hands-on experience, and walked away with new insights that will aid in their personal and professional journeys. The Training and Placement Cell looks forward to organize more events like this to continue supporting student development and career preparation.

- Sargundeep Singh MCA (M) -I

Mock Placement Drive by Career Cafe

The Training and Placement Cell successfully organized a Mock Placement Drive on February 13, 2025, in collaboration with Career Cafe. The event aimed to provide students with a realistic hands on experience of the placement process. Held at the DCSA premises, the drive commenced at 9:00 AM, with students adhering to formal dress codes and specific guidelines. The event was structured into several rounds to mimic a real placement process: a Pre - Placement Talk, followed by an MCQ Round, a Coding Round, and Mock Interviews.

The Pre-Placement Talk was delivered by Amit Singh, founder of Career Cafe and a mentor to over 1,000 students. Drawing from his extensive corporate experience, he offered valuable insights into resume building, interview preparation, and effective job-seeking strategies. He emphasized that technical expertise alone is not enough—strong communication skills, problem-solving abilities, and confidence play a crucial role in cracking interviews. The MCQ Round assessed students' abilities

in quantitative aptitude, logical reasoning, critical thinking, English grammar, and technical knowledge.Following this, the Coding Round, conducted on the HackerRank platform, tested their programming and algorithmic skills.The results of the drive were announced on February 26, 2025, recognizing the top 20 performers. Overall, the Mock Placement Drive proved to be an extremely beneficial initiative, providing students with valuable feedback and a deeper understanding of the recruitment process, ultimately helping them enhance their readiness for future job opportunities.

Ishneet Kaur MCA (M) -I

Expert Talk on Enhancing Employability in AI-Driven IT Sector

The Training and Placement Cell successfully organized an expert talk on "Upgrading Employability Skills for an AI-equipped IT Sector" on February 27, 2025. The session featured Mr. Om Kanoujia, Founder and CEO of Alar Solutions, LLP, who shared valuable insights into the rapidly evolving IT industry and the essential skills needed to succeed in an AI-driven job market.

The session aimed to enhance students' employability by helping them understand industry expectations, emerging technologies, and career opportunities in the IT sector. Mr. Kanoujia emphasized the significance of continuous learning, adaptability, and AI-related skills to remain competitive.

The event was organized under the guidance of faculty members Dr. Anuj Kumar and Dr. Kavita Taneja, who played a crucial role in organizing and facilitating the session.



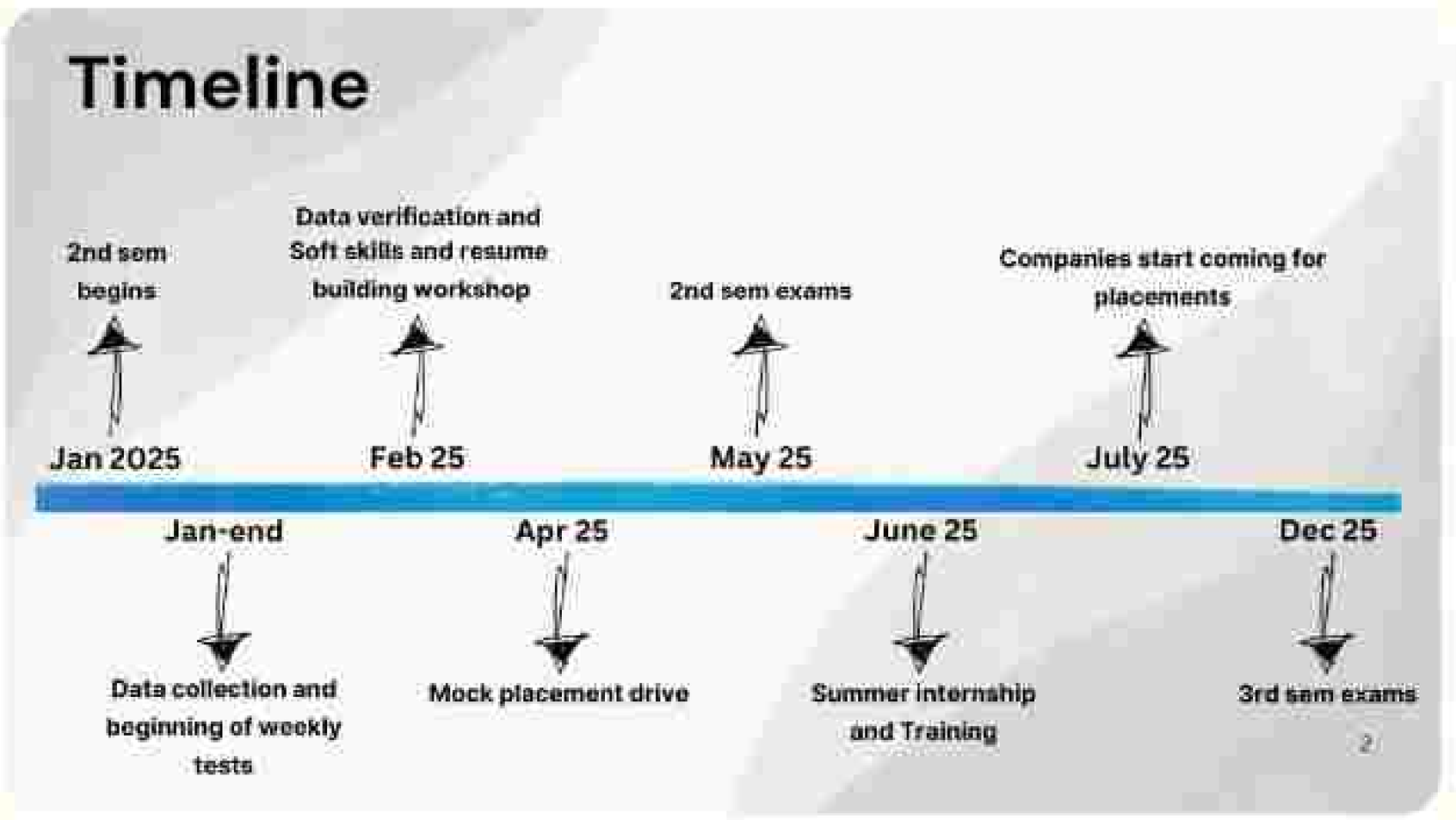
Their efforts ensured a seamless and engaging experience for the students.

The interactive discussion allowed students to engage directly with the expert, ask questions, and gain practical knowledge on career growth and industry trends. The event was highly informative and inspiring, encouraging students to embrace AI advancements and develop future-ready skills for the ever-evolving tech landscape.

-Kashish Dhamija MCA (M) -I

Essential Placement Skills: From DSA to Communication

The following insights are shared by seniors who have gone through the placement process and faced its challenges. These experiences are meant to help guide you through your own placement journey. According to their experiences, time management and planning are crucial.



Breaking down the syllabus into manageable chunks, with a focus on foundational topics like Data Structures and Algorithms (DSA), is key. Additionally, understanding concepts of OOP, C++, Java, SQL, and DBMS is essential. A deep understanding of these concepts is necessary to solve complex problems effectively.

The placement process can be mentally challenging, with different companies having varying expectations. Technical tests require strong time management, and effective communication skills are crucial in interviews. Quantitative aptitude can be challenging, but consistent practice will improve both speed and accuracy. Additionally, being able to clearly explain your projects and technical knowledge can be helpful for interviews.

For both technical and HR rounds, revising core topics and being aware about your projects is important. Researching the company in advance helps align your answers with its values and expectations. The seniors also recommend practicing coding under time constraints by participating in coding competitions and mock interviews. This builds problem-solving skills and boosts confidence.

Having strong projects on your resume is essential. Projects demonstrate your initiative and technical abilities, and participating in hackathons further enhances your resume. Focus on strengthening your technical skills, engaging in regular coding practice, and thoroughly preparing for both rounds of interviews. For preparation, online platforms are excellent resources.

For juniors, the seniors advise using the summer break to focus on DSA and coding practice. A solid project on your resume, along with hackathon experience, can give you a competitive edge. Stay consistent, learn from each experience, and don't be discouraged by setbacks.

- Ishneet Kaur MCA-1(M)

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Mr. Mangat Ram Honoured by Panjab University for Outstanding Service

On January 26, 2025, Mr. Mangat Ram, a dedicated clerk at the Department of Computer Science and Applications (DCSA), was honoured by Panjab University, Chandigarh, for his outstanding performance and dedication towards his duties. His unwavering commitment and exceptional contributions to the department's administrative efficiency have earned him this well-deserved recognition.



Mr. Mangat Ram's hard work has played a vital role in ensuring the smooth functioning of the department, making him a valued member of the DCSA community. This award, presented on Republic Day 2025, serves as a testament to his diligence and sets an inspiring example for others in the institution

- Sargundeep Singh MCA (M) -I

An expert talk on Semantic Analysis by Dr. Jasleen Kaur Bains

On February 8, 2025, Dr. Jasleen Kaur Bains delivered an expert talk on the topic "Semantic Analysis: Pre-processing Text and Implementation" at the Skill Enhancement Course (SEC) titled "From Data Science to Intelligence: Python for AI in Research". The event was hosted by the Centre for Skill Development & Entrepreneurship (CSDE) at Panjab University.



The session was aimed at research scholars and faculty members from various disciplines. She introduced the fundamental concepts of semantic analysis and explained the various text pre-processing techniques essential for effective semantic analysis.

A detailed case study was shared to illustrate these concepts in action, helping the audience grasp how these techniques are implemented practically. The talk generated significant interest, with the audience and organizers appreciating the depth of the content shared. The event fostered engaging discussions, offering valuable insights into the integration of semantic analysis in AI-driven research.

- Arsh Divya MCA (M) -I

Talk on Cyber Crime and Security at Chandigarh University



On February 21, 2025, Dr. Anuj Kumar delivered an expert talk on 'Cyber Crime and Security' at Chandigarh University, Punjab, highlighting the increasing risks of cyber crimes and the need for preventive measures. He discussed various cyber threats, security measures, and legal frameworks to combat cyber crimes, highlighting the growing importance of protecting digital systems.

He also stressed the importance of using secure payment gateways and protecting personal data on social media. Dr. Kumar also provided insights into India's IT Act 2000 and the POCSO Act, explaining their role in cyber crime prevention.

In conclusion, he urged individuals to stay informed, practice ethical online behavior, and adopt proactive cybersecurity measures to safeguard digital systems.

- Nancy MCA (E) -I

Dr. Anuj Sharma's session on Cyber Security and Safety



On February 22, 2025, Dr. Anuj Sharma delivered an insightful lecture on "Cyber Security and Safety" in Kurukshetra as part of the Capacity Building Program organized by the Central Board of Secondary Education (CBSE).

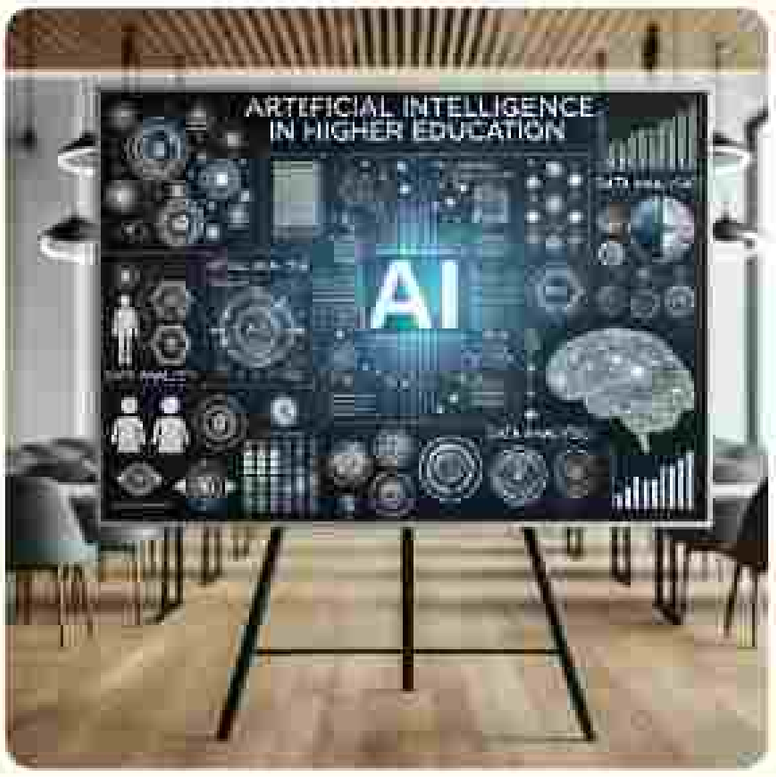
The session aimed to enhance the understanding of cyber security among educators and administrators, equipping them with essential knowledge and practical strategies to combat cyber threats and promote a secure digital environment.

The lecture encouraged individuals to adopt secure online practices, stay informed about emerging cyber risks, and take proactive measures to safeguard personal and professional data.

- Arsh Divya MCA (M) -I

Short-Term Course on AI in Higher Education

Panjab University, Chandigarh, recently hosted a short-term course on Artificial Intelligence in Higher Education from February 19 to February 26, 2025. The program aimed at exploring the transformative role of AI in academia. Organized by Prof. (Dr.) Sonal Chawla from the Department of Computer Science and Applications, in collaboration with UGC MMTTC (Malaviya Mission Teacher Training Centre), Chandigarh, the course attracted over 70 faculty members, researchers, and students from various institutions across the country.



Throughout the week-long program, participants engaged in sessions covering AI applications in teaching and learning, academic research, personalized education, and administrative automation. The course also addressed ethical considerations in AI adoption and included hands-on training and real-world case studies. Experts highlighted how AI can revolutionize education by enhancing learning experiences, optimizing administrative processes, and fostering data-driven decision-making.

The initiative was well-received, with participants appreciating the practical insights and interactive discussions. The course successfully concluded on February 26, 2025, with a certificate distribution ceremony, reinforcing the importance of AI in shaping the future of higher education.

Arsh Divya & Jaskaran Singh MCA (M) -I

Guest Lecture by Prof. Sonal Chawla on "AI to Generative AI"

On February 28, 2025, Sri Guru Gobind Singh College, Sector 26, Chandigarh, organized an IT Fest where Prof. (Dr.) Sonal Chawla, Panjab University, Chandigarh was invited as a guest speaker. The event, themed "Empowering Indian Youth for Global Leadership in Science & Innovation for Viksit Bharat", focused on AI advancements. Prof. (Dr.) Sonal Chawla discussed "AI to generative AI", emphasizing its role in daily life through applications. She highlighted AI's impact



on industries, automation, and content creation. Students actively participated in discussions on AI's future. An interactive Q&A session further enhanced understanding, making the event highly informative. The session provided valuable insights into AI's role in shaping India's technological future and its potential for innovation. The event successfully engaged students and faculty, inspiring young minds to explore AI-driven opportunities for global leadership in science and technology.

- Sargundeep Singh MCA (M) -I

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Unshaken, Unbroken, Unbound



I used to search, to chase, to plead,
Thinking love was what I'd need.
But in the quiet, I have found,
I am whole—safe and sound.

No fleeting hands, no distant gaze,
Can shape my worth or light my days.
For in my soul, a fire glows,
A love that only deeper grows.

No matter who may stay or leave,
I have myself, and I believe.
Through every storm, through every test,
I stand with me—I know me best.

I am enough, strong and free,
A world of love exists in me.
No need to beg, no need to bend,
For I am my own dearest friend.

I laugh with me, I heal my pain,
I dance alone and feel no chain.
In my own arms, I rise, I stand,
Holding my heart in my own hand.

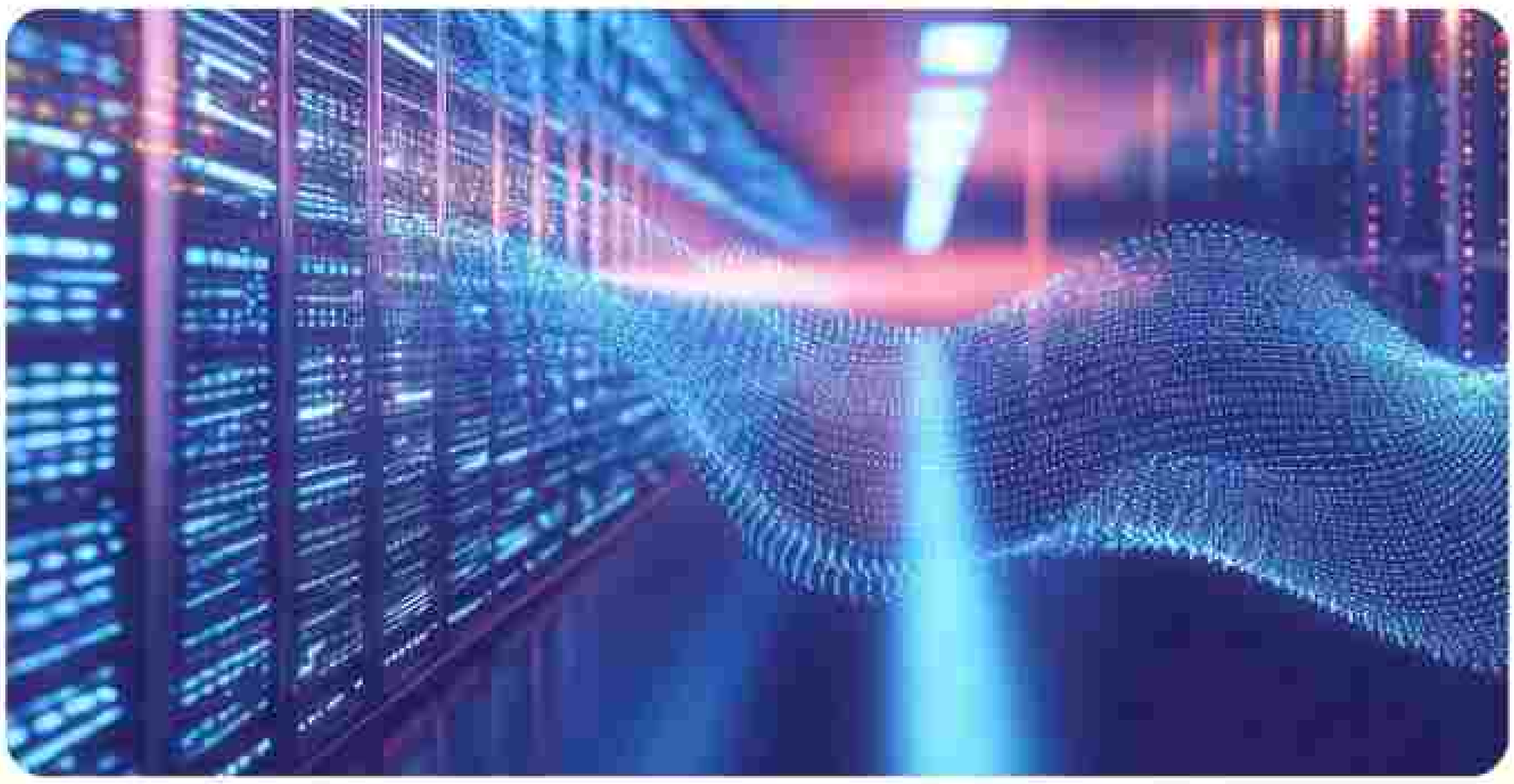
I need no mirror to see my light,
It shines in me, steady and bright.
Through every shadow,
through every doubt,
I am the love I dream about.

With every breath, I stand so tall,
Knowing I have it all.
Not in another, not in a dream,
But in my soul—a steady beam.

Let the world wander as it may,
I have found my peace to stay.
For in myself, I've come to see,
I am enough—I am complete.

- Jaskaran Singh MCA-I (M)

Invisible Threads, Unseen Ties



In the air, where silence hums,
A world of data softly drums.
No longer bound by walls or wires,
It flows like wind, ignites like fires.

Once anchored down in servers deep,
Now through the clouds it dares to leap.
A stream of bits that never cease,
A dance of data, wild and free.

The sky is filled with neon light,
As numbers drift, both day and night.
A web of code that fills the air,
Invisible yet always there.

5G whispers, swift as breeze,
And AI learns with gentle ease.
From cloud to edge, the data roams,
Connecting hearts, creating homes.

The IoT, it pulses bright,
With every step, a new insight.
Machines and minds, they intertwine,
In this vast digital design.

But though it floats in endless flight,
We guard it close, with all our might.
For in this world of endless streams,
Security must shape our dreams.

In every byte, a future lies,
Beyond the screens, beneath the skies.
For data isn't bound to stay—
It lives and breathes in light of day.

So as we look to skies anew,
The air is filled with all we do.
For in the space where clouds once were,
Data hums and lights the air.

- Diksha MCA-I (M)

Panjab University Inter College Championship

Samridhi Thakur, a student of MCA I (E), has brought pride to the institution through her outstanding achievements in Martial Arts (Pencak Silat). Demonstrating exceptional skill and dedication, she secured two silver medals in the Inter-College Championship and a bronze medal in the Khelo India Women's League (North Zone).

Samridhi's journey in sports has been a defining aspect of her life, instilling in her the values of discipline, resilience, and teamwork. Her commitment to excellence in martial arts serves as an inspiration to her peers, highlighting the importance of perseverance and hard work in achieving success.





An exceptional performance was delivered at the Inter-College Sepak Takraw Tournament, where Nikhil-I (M) secured three positions in two events, bringing recognition to Panjab University. His dedication and perseverance were evident throughout the competition, showcasing remarkable skill and sportsmanship.

Sharing his thoughts on the experience, he expressed, "It was a whole new experience for me. I have been playing this sport for a long time, but representing Panjab University was a dream come true."

This achievement highlights the significance of commitment and hard work in both academics and sports.

- Arsh Divya & Nancy MCA-I

DCSA Students Excel in UGC-NET December 2024 Examinations







Students from Panjab University's MCA program have achieved significant milestones in their academic and professional journeys.

Smriti Bansal, an MCA graduate from the batch of 2024, qualified for the JRF, UGC-NET in Computer Science, fulfilling her aspiration to join the teaching community and contribute to society. Inspired by her mentors, especially Dr. Supreet Thapar, she credits her academic success to their guidance.

Sourabh Jangra, a graduate of the MCA program from the Class of 2022-2024, has qualified for the UGC-NET in Computer Science & Applications, making him eligible to pursue a Ph.D. He described the preparation as a challenging yet fulfilling journey requiring dedication, deep knowledge, and strategic planning. This achievement has further fueled his passion for academics and research.

Jaskaran Singh, a student of MCA-I (M) Batch 2024-26, successfully cleared the UGC-NET, qualifying for both Assistant Professor and Ph.D. reflecting on his experience, he expressed surprise and joy at his achievement, highlighting the role of self-belief and consistent effort.

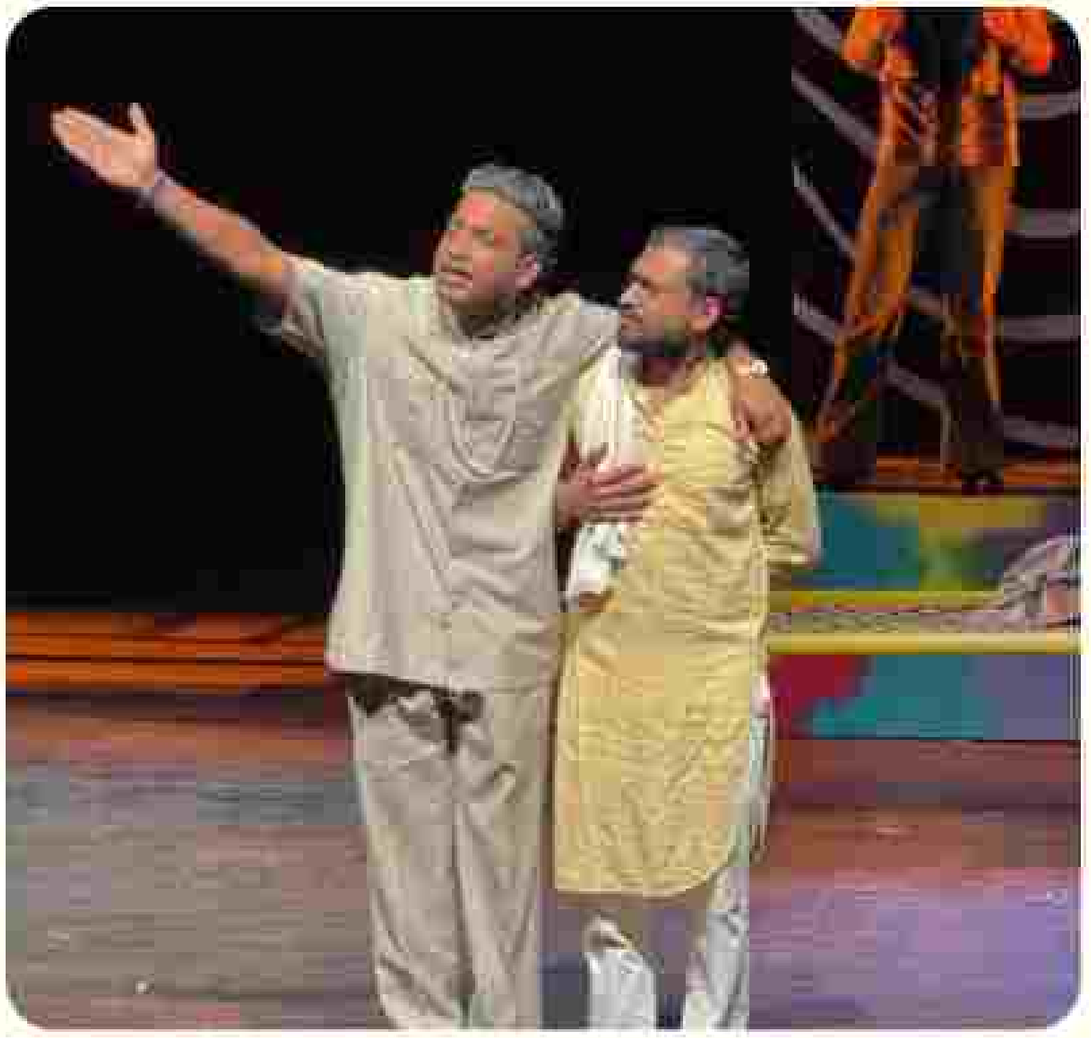
These achievements underscore the university's dedication to academic excellence and students' success, serving as a source of inspiration for future researchers and educators in the field of Computer Science

- Sargundeep Singh MCA (M) -I

Performing Beyond Limits: Lovesh's Journey of Art and Expression

On January 3, 2025, Lovesh, a student of MCA 1 (Evening), won The Common Man Show, a national-level reality show released on YouTube, securing a cash reward of ₹32,000. The competition, known for its intense format and nationwide participation, featured some of the best emerging talents in stand-up comedy and storytelling.

On February 3, 2025, his theatre production, "Madhubala – The Real Story," was selected from over 60 universities for Advitiya!, an event under Bharat Rang Mahotsav, organized by the National School of Drama (NSD),Delhi. The play, which explored the untold aspects of legendary actress Madhubala's life, was performed to a houseful audience at the prestigious Abhimanch Auditorium, NSD Delhi.



The production received critical acclaim for its powerful storytelling and performances, securing third place along with a cash prize of ₹10,000. The team was honored by the esteemed actor Pankaj Tripathi, adding to the significance of the achievement.

On February 21, 2025, Lovesh continued his winning streak, securing second position in two competitions:

- The Open Mic competition (CYANIDE), organized by University Institute of Chemical Engineering and Technology (UICET).
- The Comedy Hunt (GOONJ), organized by University Institute of Engineering and Technology (UIET).

Reflecting on his journey, Lovesh shared, "My experience of exploring and learning new things has been amazing so far. I would like to thank every single person who has been a part of my journey. Here's to achieving more milestones ahead!"

- Jaskaran Singh MCA (M) -I

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Learning Hub

Tech Updates of the Month

AI Action Summit 2025
Global leaders convened in Paris from February 6-11, co-chaired by France and India, pledging €200 billion for AI advancements.



Samsung unveiled the Galaxy S25 series on February 7, featuring advanced AI capabilities and enhanced camera technology.



Nvidia GeForce RTX 50 Series
Nvidia introduced the Blackwell GPUs, redefining AI-driven gaming with superior performance.



IM-2 Lunar Mission
Intuitive Machines launched the IM-2 mission, aiming to explore lunar water ice using advanced lander technology.



Nvidia launches RTX 5090 GPU with AI-driven enhancements, improving graphics performance for gaming and computing.



Microsoft's Majorana 1 Chip
Microsoft unveiled a topological quantum processor, the Majorana 1, marking a breakthrough in quantum computing.



Toyota introduced an electric vehicle featuring a 745-mile range and 10-minute charging, powered by solid-state batteries.



NASA's latest Mars rover, equipped with AI, enhances autonomous exploration capabilities on the Martian surface.

Google Cloud AI Enhancements:

Google Cloud AI enhancements offer powerful tools that make it easier for businesses, developers, and researchers to integrate artificial intelligence and machine learning into their operations. A key feature is AutoML, which allows users with little AI expertise to create custom models tailored to their specific needs. This automation simplifies the complex process of model creation, making AI more accessible and accelerating innovation.

Additionally, Google Cloud provides pre-trained models for various applications, such as Cloud Vision API for image recognition and Cloud Natural Language API for text analysis. These pre-built models save time and resources by offering ready-to-use solutions for tasks like object detection, sentiment analysis, and language translation.

For those looking to build custom models, Google Cloud offers robust AI infrastructure, including access to TPUs (Tensor Processing Units) and GPUs, which significantly speed up model training. This infrastructure is scalable, allowing users to manage both small experiments and large-scale AI workloads efficiently.

Furthermore, Google Cloud emphasizes AI ethics and fairness, offering tools like the What-If Tool, which helps developers evaluate and mitigate biases in their models. This ensures that AI systems are not only effective but also fair and transparent.

With these tools and advancements, Google Cloud AI supports industries such as healthcare, finance, retail, and manufacturing, enabling businesses to improve efficiency, decision-making, and customer experiences. Overall, Google Cloud AI provides a comprehensive platform that empowers users at all levels to harness the full potential of AI while maintaining ethical standards.

-Sargundeep MCA-I (M) & Ishneet MCA-I (M)

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Dr. Abhinav Kathuria:

"A Journey of Excellence in Research and Education"



Dr. Abhinav Kathuria, an esteemed alumni of the Department of Computer Science and Applications (DCSA), pursued his MCA and PhD in Computer Science (2011-2024). Currently an Assistant Professor at R.S.D. College, Ferozepur, continues to inspire students in academia and research.

Reflecting on his time at DCSA, he describes it as a transformative phase that shaped his career and personal growth. He encourages students to seize opportunities, collaborate, and embrace lifelong learning, believing these skills will prepare them for the future. His cherished memories at DCSA include engaging classroom discussions, coding sessions, collaborative projects, and vibrant seminars, which instilled perseverance, curiosity, and teamwork.

Dr. Kathuria has made significant research contributions, with publications in reputed journals such as SN Computer Science and Journal of Circuits, Systems and Computers. He has presented research at esteemed conferences, including the International Conference on Computational Methods and Data Engineering (Springer, 2020) and CHASCON-2018 at Panjab University. He also cleared UGC Computer Science for JRF & NET (2016). In recognition of his achievements, he received the Young Achiever Award by PSM Rashtriya Shiksha Ratan Sanman in 2024.

As an educator, he is committed to mentoring students, contributing to curriculum development, and serving on the Undergraduate and Postgraduate Boards of Studies at Panjab University. His journey reflects dedication, perseverance, and a passion for academia, inspiring students to excel and innovate in their careers.

- Arsh Divya MCA-I(M)

A Successful Ph.D Defense By Dr. Balwinder Kaur



Dr. Balwinder Kaur successfully defended her Ph.D. thesis on January 31, 2025. Title of her thesis was "A Multi-Classifier Framework for Academic Performance Prediction of Students", which focuses on predicting student performance using Educational Data Mining (EDM) and Machine Learning (ML) techniques. Her research utilizes student data from Panjab University, Chandigarh, incorporating data preprocessing, feature selection, and Synthetic Minority Over-sampling Technique (SMOTE) to handle dataset imbalances.

The study evaluates various classifiers, including Decision Trees, Bayesian Classifiers, k-NN, ANN, and SVM, and explores ensemble learning methods. It introduces HE2MAP (Heterogeneous Ensemble to Measure Academic Performance), integrating J48, JRip, and MLP classifiers, achieving 99.39% accuracy on training data and 97% on test data, outperforming existing models.

Dr. Balwinder Kaur's research significantly contributes to educational analytics and institutional decision-making, emphasizing proactive strategies for academic success. The dataset and source code will be publicly available to encourage further research and development in student performance prediction.

- Arsh Divya MCA-I(M)

A Successful Ph.D. Defense By Dr. Supreet Thapar



On January 31, 2025, Dr. Supreet Thapar successfully defended her Ph.D. thesis, "Design and Development of an Algorithmic Framework for Cluster-Based Cab Recommender System (CBCRS)," which addresses challenges in identifying high-demand cab pickup locations. Her research introduces CBCRS, an AI-driven framework integrating Clustering Algorithms, Collaborative Filtering, and Density Analysis to enhance recommendation accuracy. The study follows a four-stage methodology, covering framework design, development, implementation using Python and Django, and rigorous testing on Standard and Non-Standard datasets. Results show that CBCRS significantly improves recommendation accuracy and operational efficiency in the cab industry.

Dr. Supreet Thapar's research makes a valuable contribution to Computer Science, Data Science, and Transportation Analytics, and her open-source framework paves the way for further advancements in cab recommendation technologies.

- Arsh Divya MCA-I(M)

Plagiarism: A Threat to Academic Integrity

Plagiarism is frequently discussed in research writing as a form of research misconduct. Researchers may commit plagiarism either intentionally or unintentionally, violating research ethics. Research ethics comprise a set of guidelines that ensure integrity, respect, and compliance with human rights in research. Therefore, researchers have a fundamental duty to uphold these ethical principles.

Plagiarism involves incorporating the ideas or work of others, with or without the permission of the original creator or researcher in your work. It includes the use, publication, or presentation of someone else's text, data, images, illustrations, graphs, or ideas as one's own without proper attribution. It is one of the most common and serious form of academic dishonesty that undermines the credibility of a researcher's work. Various forms of plagiarism include self-plagiarism, cyber-plagiarism, word-for-word (Verbatim) plagiarism, Mosaic plagiarism (Patch Writing), Thesaurus (Paraphrase), Global Plagiarism, and Accidental Plagiarism.

In 2018, the University Grants Commission (UGC) of India introduced the "Promotion of Academic Integrity and Prevention of Plagiarism in Higher Educational Institutions) Regulations, 2018 " to prevent academic misconduct and foster a culture of original and ethical research in India's higher education system.

Different levels of plagiarism based on similarity:

- **Level 0:** Similarities up to 10% – Minor similarities; no penalty.
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4. Grammarly- https://www.grammarly.com/plagiarism-checker

Article written by : S. RATNA
(Research Scholar)

Research Trends in Face Emotion Recognition (FER)

Humans are emotional beings, doing many things in the flow of emotions. Many unwanted situations in their lives can be avoided by identifying emotions at the right time and taking proactive measures to prevent undesirable outcomes. Can some automated systems recognize human emotions from face, speech, or text?
Yes, such a system is called Face Emotion Recognition. Recent advancements in artificial intelligence have given way to specialized models that can recognize emotions. One commonly known and recently researched field is Face Emotion Recognition (FER). It is an intersection of artificial intelligence, computer vision, and human psychology. If you are a researcher looking ahead for some new and exciting research field, you may get some ideas from this discussion.

Transformer-Based Models in FER

Deep learning approaches like transformers have proven their superior feature extraction and pattern recognition abilities that apply to FER as well. Vision Transformers (ViTs) and hybrid CNN-ViT models are a lot improved in accuracy. Models like ResNet, EfficientNet, and DenseNet are nowadays used as backbone architectures for feature extraction in FER systems. CNN models are also combined with attention mechanisms to improve performance in Emotion Recognition by capturing spatial and contextual features.

Vision Transformers (ViTs) with the support of Self-Attention Mechanisms are also an efficient technique in FER systems to identify spatial relationships and global features of faces to identify changing emotions. Many researchers have used Swin Transformers for hierarchical feature extraction in complex facial emotion datasets. Graph Neural Networks (GNNs) are also being implemented in this field as structural relationships between different facial landmarks and enhanced feature extraction can be represented in a better manner. Dynamic Graph Convolutional Networks (DGCN) are in focus to capture movement-based facial expressions.

Multimodal Emotion Recognition

Multiple data sources, such as voice, physiological signals, and contextual information can enhance the process of facial emotion recognition. Integration of audio with visual Emotion Recognition is a growing research field that involves the fusion of facial expressions and speech features. Multimodal Transformers (MMTs) have been experimented to combine visual and audio data effectively.

Physiological Signals (EEG, Heart Rate) can additionally provide the required support to improve emotion classification. Text modality is another important aspect of this research where text-based sentiment analysis in conjugation with visual face expressions can lead to better social media and human-computer interactions.

Handling Occlusions, Pose Variations, and Real-World Challenges

FER systems are often challenged by issues like occlusions due to masks and other accessories, pose variations, and changing lighting conditions. Occlusion-Aware FER Models can be implemented by utilizing attention mechanisms and GAN-based inpainting after training on datasets with synthetic occlusions. 3D Morphable Models (3DMMs) support facial expression reconstruction in the presence of occlusions. Cross-domain and Few-Shot Learning with Domain Adaptation (DA) are again good research areas to generalize FER models across different datasets which is quite useful in the absence of labeled datasets.

Applications of FER Research

FER can be effectively applied for depression detection, autism spectrum disorder (ASD) analysis, and stress monitoring. Timely emotion tracking with such applications can help people with suitable therapy. AI assistants and chatbots can adapt to provide the right responses based on the emotions of users. FER is redefining gaming and virtual reality (VR) immersive experiences making them more engaging and exciting. FER systems can be a huge support for security systems, lie detector systems as well as behavioral analysis during forensic investigations. Surveillance systems can be integrated FER with anomaly detection in human behavior.

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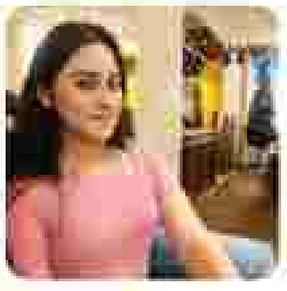
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